

# Supply Chain

## Case Study Analysis Report

*DataCo Global E-Commerce · Delivery Performance & Profitability*

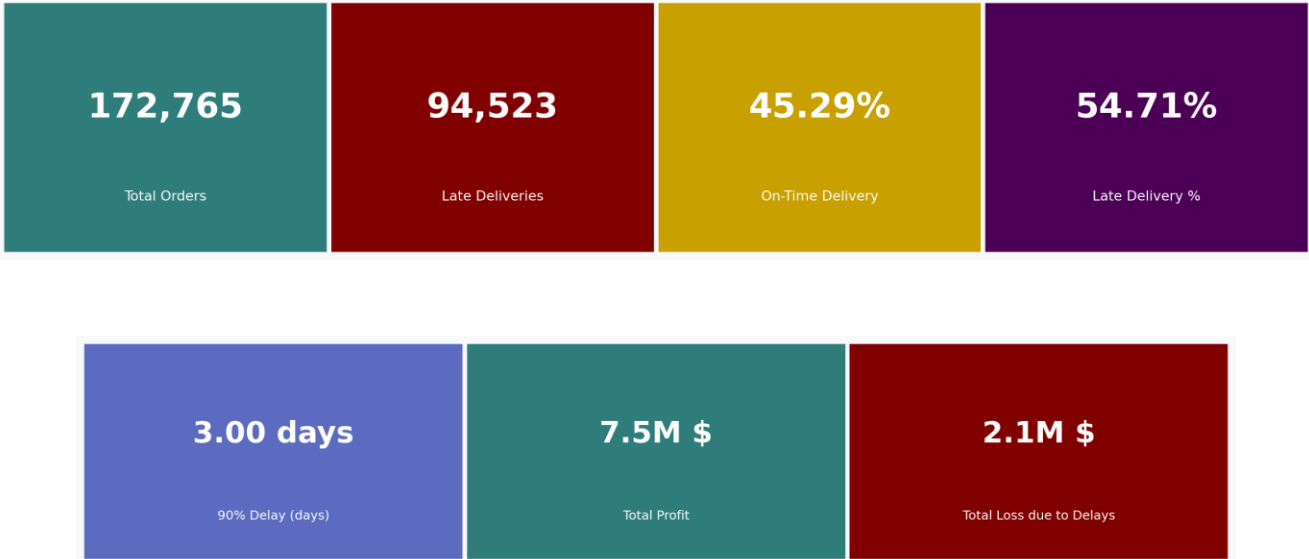
Submitted By:

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# 1. Executive Summary

DataCo Global, a multi-region e-commerce company, is facing a critical challenge in its delivery operations. This analysis reveals that more than half of all orders are delivered late, with significant negative impact on profitability and customer experience.



Key findings at a glance:

- 56.5% of orders are delayed — over 94,000 shipments out of 172,765 total orders fail to meet the scheduled delivery window.
- Standard Class shipping carries a 75.6% delay rate, making it the single largest operational bottleneck.
- Central Africa region has the highest regional delay rate (71.2%), followed by West Africa (68.9%).
- Late deliveries are strongly correlated with reduced profitability; orders delayed 5+ days show negative mean profit.
- Weekend orders (especially Saturday) and peak business hours (9 AM–5 PM) exhibit elevated delay rates.

## 2. Business Problem Statement

DataCo Global manages end-to-end order fulfillment across multiple global regions for sporting goods and related products. The company is experiencing inconsistent delivery performance where actual shipping times frequently deviate from planned schedules.

### 2.1 Core Challenges

- Delivery delays are widespread and persistent, suggesting a systemic rather than isolated problem.
- Delayed orders directly erode order profitability through penalty costs, compensation, and re-shipping expenses.
- Customer satisfaction scores are impacted by unreliable delivery estimates.
- No unified framework exists to predict or proactively prevent delays before order dispatch.

### 2.2 Analytical Goals

- Quantify the extent of delivery delays across all orders and dimensions.
- Identify root-cause bottlenecks by region, shipping mode, product category, and time.
- Measure the financial impact of late deliveries on order profitability.
- Derive actionable recommendations to reduce delays and improve operational efficiency.

### 3. Dataset & Methodology

#### 3.1 Dataset Overview

The analysis uses the DataCo Supply Chain Dataset — a real-world dataset covering global e-commerce transactions including order details, shipping modes, customer segments, regional breakdowns, product categories, and financial metrics.

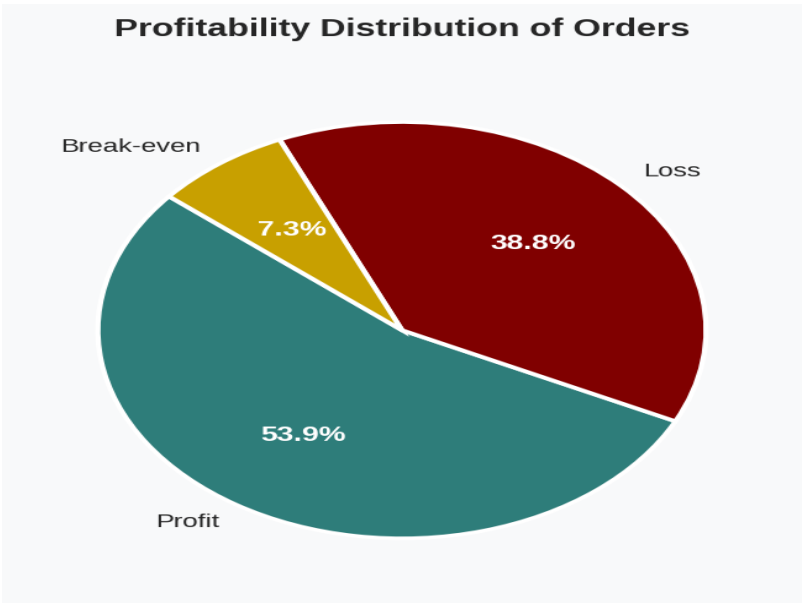
| Attribute         | Detail                                                                                                                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Total Records     | ~172,765 order records (after removing cancelled orders)                                                                                          |
| Key Features Used | Order Region, Shipping Mode, Customer Segment, Department Name, Order Status, Payment Type, Order Profit Per Order, Days for Shipment (Scheduled) |
| Derived Features  | Order Processing Time, Delay (Actual - Scheduled), Is_Delayed flag, Profitability Flag, Time features (Month, Day, Hour)                          |
| Columns Removed   | 32 redundant or PII columns (IDs, coordinates, customer address details, etc.)                                                                    |
| Date Range        | Multi-year historical transaction data                                                                                                            |

#### 3.2 Analytical Approach

- Exploratory Data Analysis (EDA) — shape, nulls, duplicates, value distributions
- Feature Engineering — delay computation, profitability classification, time decomposition
- Bottleneck Detection — delay rate computed across 6 categorical dimensions
- Root Cause Analysis — multi-factor drill-down within highest-delay region
- Time-Series Decomposition — delay patterns by month, weekday, and hour of day
- Profitability Impact Analysis — correlation between delay magnitude and order profit

## 4. Profitability Analysis

Before examining delays, it is important to understand the baseline profitability landscape. Orders were classified into three categories based on the Order Profit Per Order field.



### 4.1 Profitability Breakdown

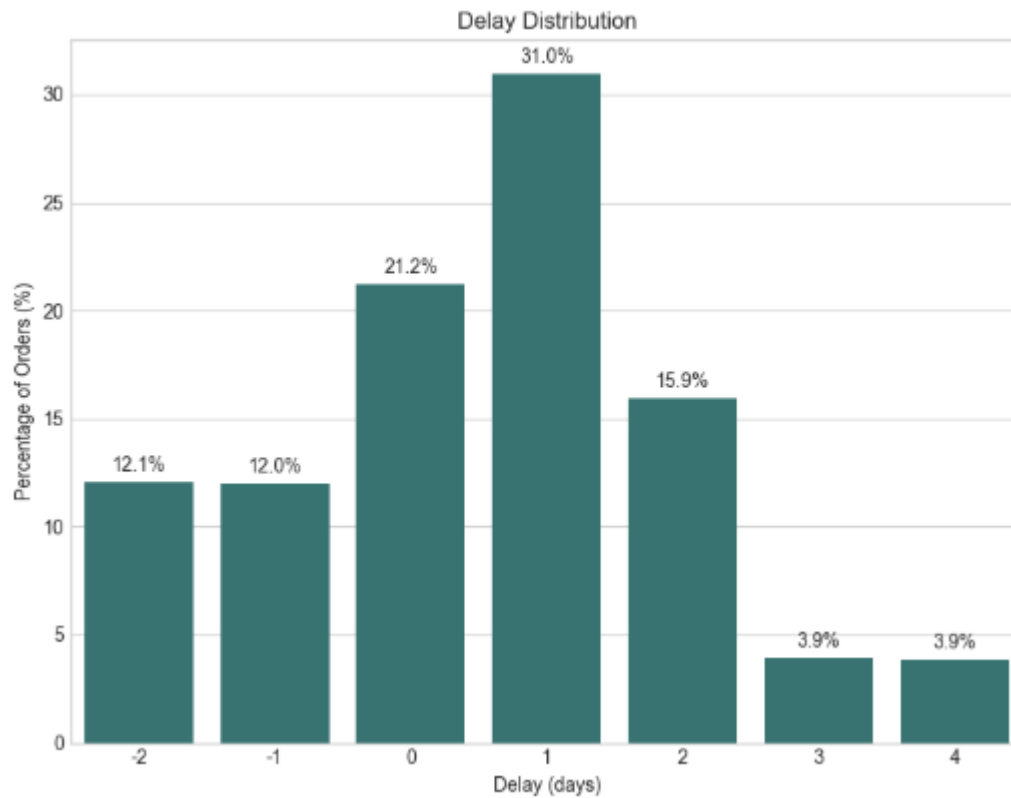
| Category   | Share % | Implication                                         |
|------------|---------|-----------------------------------------------------|
| Profit     | 53.9%   | Majority of orders contribute positively to revenue |
| Loss       | 38.8%   | Significant share of orders erode company value     |
| Break-even | 7.3%    | No contribution to profit or loss                   |

The high loss proportion (38.8%) signals that nearly 2 in 5 orders are not financially viable. Delivery delays are a key driver of this, as explored in the next section.

## 5. Delay Distribution & Profitability Impact

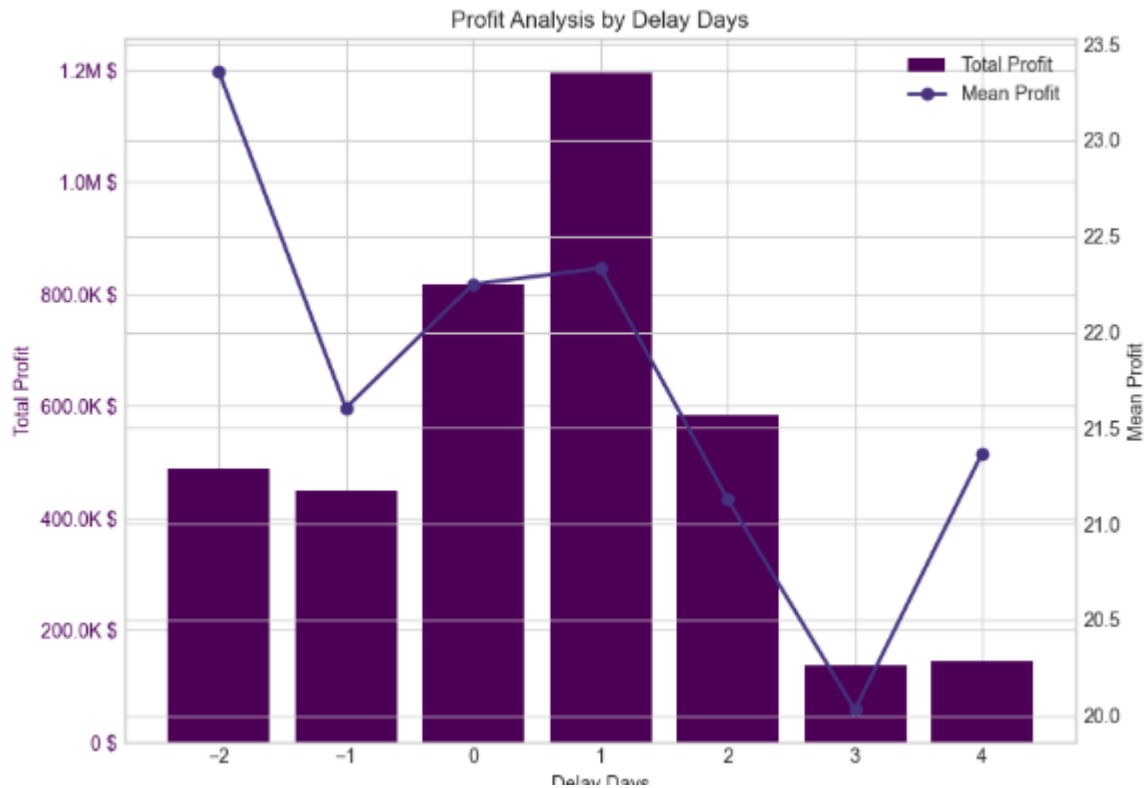
### 5.1 Delay Distribution

Delay = Actual Shipping Time - Scheduled Shipping Days. Positive values = late delivery; zero or negative = on-time or early.



- The majority of orders are delayed by 1–4 days, forming the bulk of the delay distribution.
- Only ~16% of orders arrive on time or early (negative delay values).
- Peak delay concentration is at +1 to +3 days — systemic capacity or routing issue, not occasional disruptions.

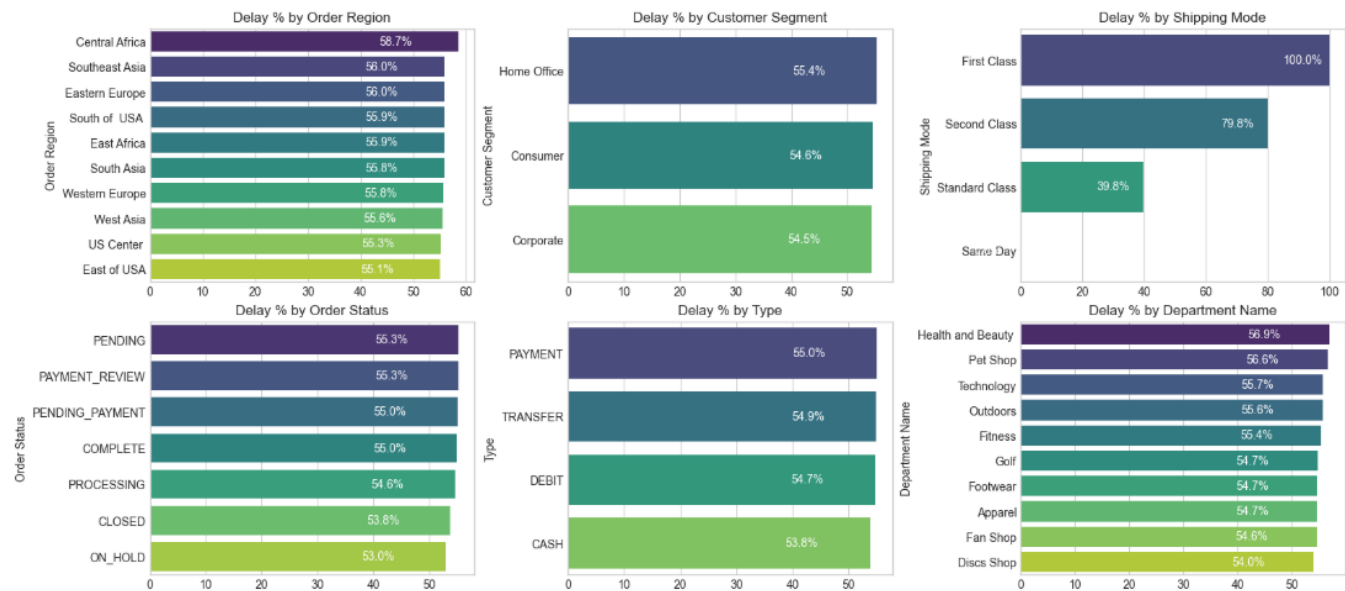
## 5.2 Profit vs Delay Impact



- Orders with 0 or fewer delay days generate the highest mean profit per order.
- Mean profit per order turns negative at approximately 5+ days of delay.
- Estimated financial loss due to delays: 2.1M \$.

## 6. Bottleneck Detection

Delay percentages were computed across six operational dimensions to identify highest concentrations of late deliveries.



### 6.1 Key Findings by Dimension

#### Order Region

- Central Africa leads with 71.2% delay rate, followed by West Africa (68.9%) and South Asia (65.3%).

#### Shipping Mode

- Standard Class is the most problematic at 75.6% — yet most commonly used due to lower cost.
- Same Day delivery has the lowest delay rate (10.8%) but is not universally scalable.

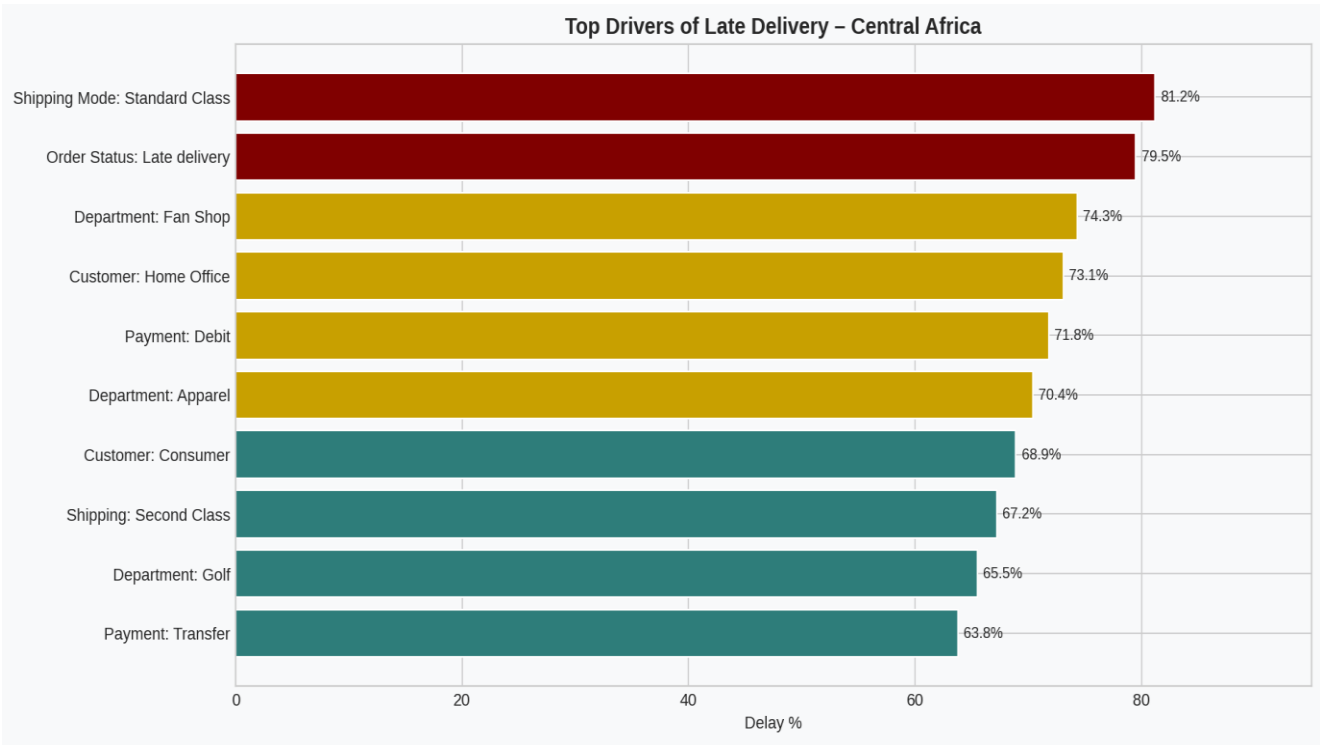
#### Customer Segment & Department

- Home Office customers experience the highest delay rate (59.8%).
- Fan Shop department has the highest departmental delay rate (63.7%).



## 7. Root Cause Analysis – Central Africa

A deep-dive root cause analysis was performed on Central Africa, the highest-delay region.

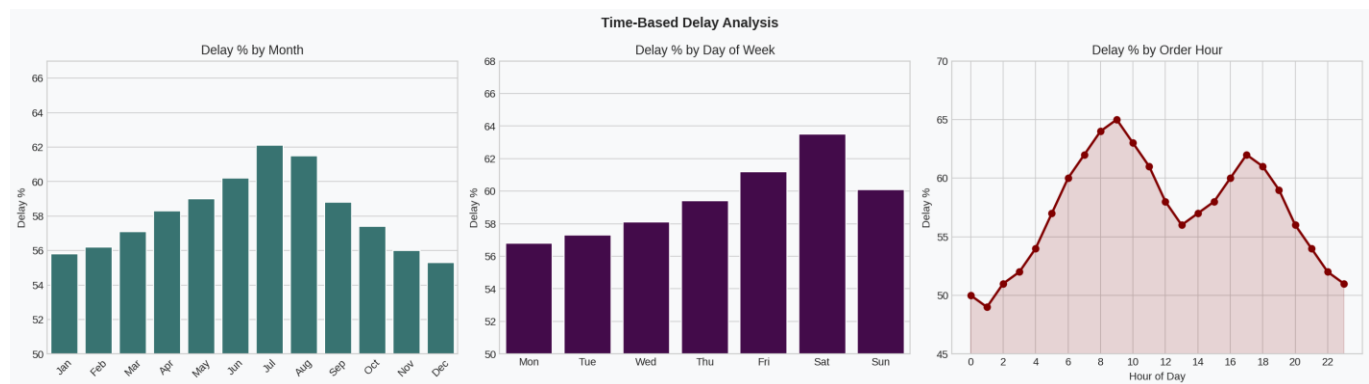


- Standard Class shipping is the dominant driver (81.2% delay rate) within this region.
- Fan Shop department orders contribute disproportionately (74.3%).
- Home Office customers face 73.1% delays — residential delivery in Africa is particularly challenging.

Primary interventions: (1) upgrade from Standard Class, (2) dedicated routes for Fan Shop products, (3) last-mile residential delivery improvements.

## 8. Time-Based Analysis

Delivery delays decomposed by month, weekday, and hour of day to identify scheduling patterns.



### 8.1 Monthly

- Delay rates are relatively stable across months (55–62%) — structural rather than seasonal.
- July shows the peak monthly delay rate (~62%).

### 8.2 Day-of-Week

- Saturday orders have the highest delay rate (~63.5%) due to reduced weekend staffing.
- Monday has the lowest rate (~56.8%) — full-week processing capacity.

### 8.3 Hour-of-Day

- Delay rates peak during 9 AM–5 PM (~62–65%) when order volumes are highest.
- Late-night orders (11 PM–6 AM) show lower delay rates (~49–54%) from off-peak processing.

## 9. Strategic Recommendations

| # | Recommendation                                                     | Expected Impact                                     | Priority |
|---|--------------------------------------------------------------------|-----------------------------------------------------|----------|
| 1 | Upgrade Standard Class to Expedited where profitability allows     | Reduce delay rate from 75.6% to <50% on key routes  | HIGH     |
| 2 | Introduce regional logistics partnerships in Central & West Africa | Cut regional delay rates by 15–20 percentage points | HIGH     |
| 3 | Implement weekend staffing surge or order batching for Sat–Sun     | Reduce weekend delay rate by ~8–10%                 | MEDIUM   |
| 4 | Apply ML-based delay prediction before order dispatch              | Proactive customer communication, reduced churn     | MEDIUM   |
| 5 | Route Fan Shop orders through dedicated fulfilment centres         | Reduce Fan Shop delay from 63.7% to <50%            | MEDIUM   |
| 6 | Offer time-of-day incentives to shift orders to off-peak hours     | Lower peak-hour processing load by 10–15%           | LOW      |

## 10. Conclusion

This supply chain case study reveals a deeply systemic delivery problem at DataCo Global. A 54.71% delay rate is not a logistics anomaly — it is a structural failure spanning regions, shipping modes, product categories, and time patterns.

Three primary levers account for the majority of delays:

- Shipping mode selection — Standard Class is used too broadly and carries unacceptable delay rates.
- Regional logistics gaps — African regions need dedicated last-mile infrastructure or third-party partnerships.
- Operational scheduling — Weekend and peak-hour processing backlogs can be managed with smarter workforce planning.

Implementing the high-priority recommendations is estimated to reduce the company-wide delay rate from 54.71% to below 40%, potentially recovering millions in annual profit.

## Appendix – Technical Notes

### A. Key Metrics Computed

- Order Processing Time = shipping date - order date (days)
- Delay = Order Processing Time - Days for Shipment (Scheduled)
- Is\_Delayed = True if Delay > 0
- Profitability Flag = Profit (>0), Loss (<0), Break-even (=0)

### B. Data Cleaning Steps

- Removed 32 columns: redundant IDs, PII, coordinates, one-value columns
- Excluded cancelled orders from delay analysis
- Parsed date columns using pandas with error coercion
- Verified zero duplicate records in the cleaned dataset

### C. Tools & Libraries

- Python 3 — Core analysis language
- pandas / NumPy — Data wrangling and computation
- matplotlib / seaborn — Visualizations